the compiler's standards.

The following Table 9.5 shows how much of memory space is allocated for each fundamental data type. Remember that, every data is stored inside the computer memory in the form of binary digits (See Unit I Chapter 2).

Table 9. 5 Memory allocation for Fundamental data types

Data type	Space in memory		Range of value
	in terms of bytes	in terms of bits	
char	1 byte	8 bits	-128 to 127
int	2 bytes	16 bits	-32,768 to 32,767
float	4 bytes	32 bits	3.4×10^{-38} to 3.4×10^{38} -1
double	8 bytes	64 bits	1.7×10^{-308} to 1.7×10^{308}-1

9.12.3 Data type modifiers:

Modifiers are used to modify the storing capacity of a fundamental data type except void type. Usually, every fundamental data type has a fixed range of values to store data items in memory. For example, int data type can store only two bytes of data. In reality, some integer data may have more length and may need more space in memory. In this situation, we should modify the memory space to accommodate large integer values. **Modifiers can be used to modify (expand or reduce) the memory allocation of any fundamental data type. They are also called as Qualifiers.**

There are four modifiers used in C++. They are:

(1) signed (2) unsigned (3) long (4) short

These four modifiers can be used with any fundamental data type. The following Table 9.6 shows the memory allocation for each data type with and without modifiers.

Integer type

Table 9.6 Memory allocation for Data types

Data type		Space in memory		Range of value
		in terms of bytes	in terms of bits	
short	short is a short name for **short int**	2 bytes	16 bits	-32,768 to 32,767
unsigned short	an integer number without minus sign.	2 bytes	16 bits	0 to 65535
signed short	An integer number with minus sign	2 bytes	16 bits	-32,768 to 32,767
Both short and signed short are similar				
int	An integer may or may not be signed	2 bytes	16 bits	-32,768 to 32,767

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